

Bengaluru Ride Data Analysis

End-to-End Data Analytics Project

Python · SQL · Power BI · ChatGPT

100K

Total Records

62%

Success Rate

7

Vehicle Types

50

Pickup Zones

4.0

Avg Rating

1. Project Overview

This report details an end-to-end data analytics project focused on ride booking data from Bengaluru, India. The initiative simulated a real-world ride-hailing platform's operations to generate comprehensive insights into customer behavior, vehicle performance, and operational efficiency. Leveraging a suite of modern data tools, the project covered data generation, cleaning, analysis, and visualization, culminating in an interactive dashboard designed for business intelligence.

Tools & Technologies

ChatGPT	Synthetic generation of a large-scale, realistic 100,000-record dataset
Python (Pandas, NumPy)	Data manipulation, cleaning, and preprocessing
SQL	Querying the cleaned dataset to extract specific business insights
Power BI	Developing an interactive dashboard to visualize key metrics and trends

2. Dataset Generation & Specifications

A foundational step in this project was the creation of a realistic synthetic dataset using ChatGPT, engineered to produce a comprehensive transactional log of 100,000 records for Bengaluru city, detailing various aspects of ride bookings with an overall success rate of 62%.

Dataset Specifications

Metric	Value
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Total Records	100,000
Time Period Covered	1 Month
City of Focus	Bengaluru, India
Distinct Vehicle Types	7
Distinct Pickup Locations	50
Overall Success Rate	62%
Total Columns	19

3. Data Cleaning & Preprocessing

Prior to any analytical work, the raw synthetic dataset underwent rigorous cleaning and preprocessing using Python. This phase was crucial to ensure data quality, consistency, and reliability for subsequent analysis.

- Data Loading into Pandas DataFrame for efficient manipulation
- Data Type Verification — corrected types for numeric and categorical fields
- Duplicate Record Removal — 0 duplicates found, maintaining full data integrity
- Column Name Standardization using snake_case convention
- Missing Value Identification and impact analysis
- Data Quality Verification — logical consistency checks across all fields

100,000	0	✓ Standardized
Records Verified	Duplicates Found	Column Names

4. SQL Analysis

The cleaned dataset was subjected to SQL analysis to answer ten specific business questions. Each query was designed to extract meaningful, actionable insight from the structured ride data.

No.	Business Question
Q1	Retrieve all successful bookings
Q2	Find the average ride distance for each vehicle type
Q3	Get the total number of cancelled rides by customers
Q4	List the top 5 customers who booked the highest number of rides

Q5	Get rides cancelled by drivers due to personal and car-related issues
Q6	Find max and min driver ratings for Prime Sedan bookings
Q7	Retrieve all rides where drop location is Indiranagar
Q8	Find the average customer rating per vehicle type
Q9	Calculate the total booking value of successfully completed rides
Q10	List all incomplete rides along with their reason

■ Snapshot: SQL Queries — Part 1 (Q1 – Q5)

Business questions 1 through 5 executed in the SQL editor

```
1  -- 1. Retrieve all successful bookings:
2  • SELECT booking_id, booking_status
3    FROM ride_data
4    WHERE booking_status = "Success"
5  -- 2. Find the average ride distance for each vehicle type:
6  ✖ SELECT vehicle_type,
7      AVG(ride_distance) AS avg_ride_distance
8    FROM ride_data
9    GROUP BY vehicle_type;
10
11
12  -- 3. Get the total number of cancelled rides by customers:
13
14  • select count(*) FROM ride_data where booking_status = "Cancelled by Customer";
15
16
17  -- 4. List the top 5 customers who booked the highest number of rides:
18
19  • SELECT customer_id,
20      COUNT(booking_id) AS total_rides
21    FROM ride_data
22    GROUP BY customer_id
23    ORDER BY total_rides DESC
24    LIMIT 5;
25
26
27  -- 5. Get the number of rides cancelled by drivers due to personal and car-related issues:
28
29  • SELECT COUNT(*) FROM ride_data WHERE cancelled_rides_by_driver = 'Personal & Car
30    related issue';
```

Figure 1 — SQL editor showing queries Q1–Q5: successful bookings, average ride distance per vehicle type, cancelled rides count, top 5 customers by rides, and driver-cancelled rides by reason.

■ Snapshot: SQL Queries — Part 2 (Q6 – Q10)

Business questions 6 through 10 executed in the SQL editor

```
31
32  -- 6. Find the maximum and minimum driver ratings for Prime Sedan bookings:
33  • SELECT MIN(driver_ratings) as min_rating, MAX(driver_ratings) as max_rating
34    FROM ride_data
35    WHERE vehicle_type = "Prime Sedan" AND booking_status = "Success";
36  -- 7. Retrieve all rides where drop location is indiranagar:
37
38  • SELECT * FROM ride_data WHERE drop_location = "indiranagar";
39
40  -- 8. Find the average customer rating per vehicle type:
41  • SELECT vehicle_type, avg(customer_rating) as avg_cstm_rating
42    FROM ride_data
43    GROUP BY vehicle_type;
44  -- 9. Calculate the total booking value of rides completed successfully:
45  • SELECT sum(booking_value) as "total booking value" FROM ride_data WHERE incomplete_rides = 0;
46  -- 10. List all incomplete rides along with the reason:
47  • SELECT booking_id,
48      customer_id,
49      vehicle_type,
50      incomplete_rides_reason
51    FROM ride_data
52    WHERE incomplete_rides = 0;
53  --
```

Figure 2 — SQL editor showing queries Q6–Q10: driver rating ranges for Prime Sedan, Indiranagar drop rides, average customer rating by vehicle, total successful booking value, and incomplete rides with reasons.

5. Key Findings & Business Insights

Overall Booking Performance

- 62% of all bookings were successfully completed over the analysis period
- 38% of bookings were unsuccessful, highlighting cancellation and incomplete ride issues
- Customer and driver cancellations together account for the majority of failed bookings

Customer Behavior

- Top customers by ride frequency are strong candidates for loyalty programs
- Customer cancellations are often linked to wait times, pricing, or changed plans
- The top 5 customers demonstrate significantly higher engagement than average users

Vehicle Performance

- Prime Sedan and Prime SUV generate higher revenue per ride
- Bike rides had the highest booking volume, reflecting urban mobility patterns
- Average ride distances vary meaningfully across vehicle categories

Ratings Analysis

- Average driver rating: 4.0 — indicating consistently satisfactory service quality
- Average customer rating: 4.01 — stable across all vehicle categories
- Prime Sedan driver ratings showed the widest min-max spread among all types

Cancellation Insights

- Top reasons: Change of plans, Driver not moving towards pickup, Wrong address
- Driver-side cancellations due to personal and car issues are a measurable pattern
- Addressing navigation and ETA accuracy could significantly reduce cancellations

Location Intelligence

- Indiranagar, Bellandur, and Whitefield are the highest-demand drop zones
- Top pickup zones align with Bengaluru's major commercial and residential hubs
- Dynamic pricing and driver allocation can be optimized by zone-level demand data

■ Snapshot: Power BI Dashboard — Bengaluru Ride Data Analysis

Interactive business intelligence dashboard showing key KPIs and visualizations

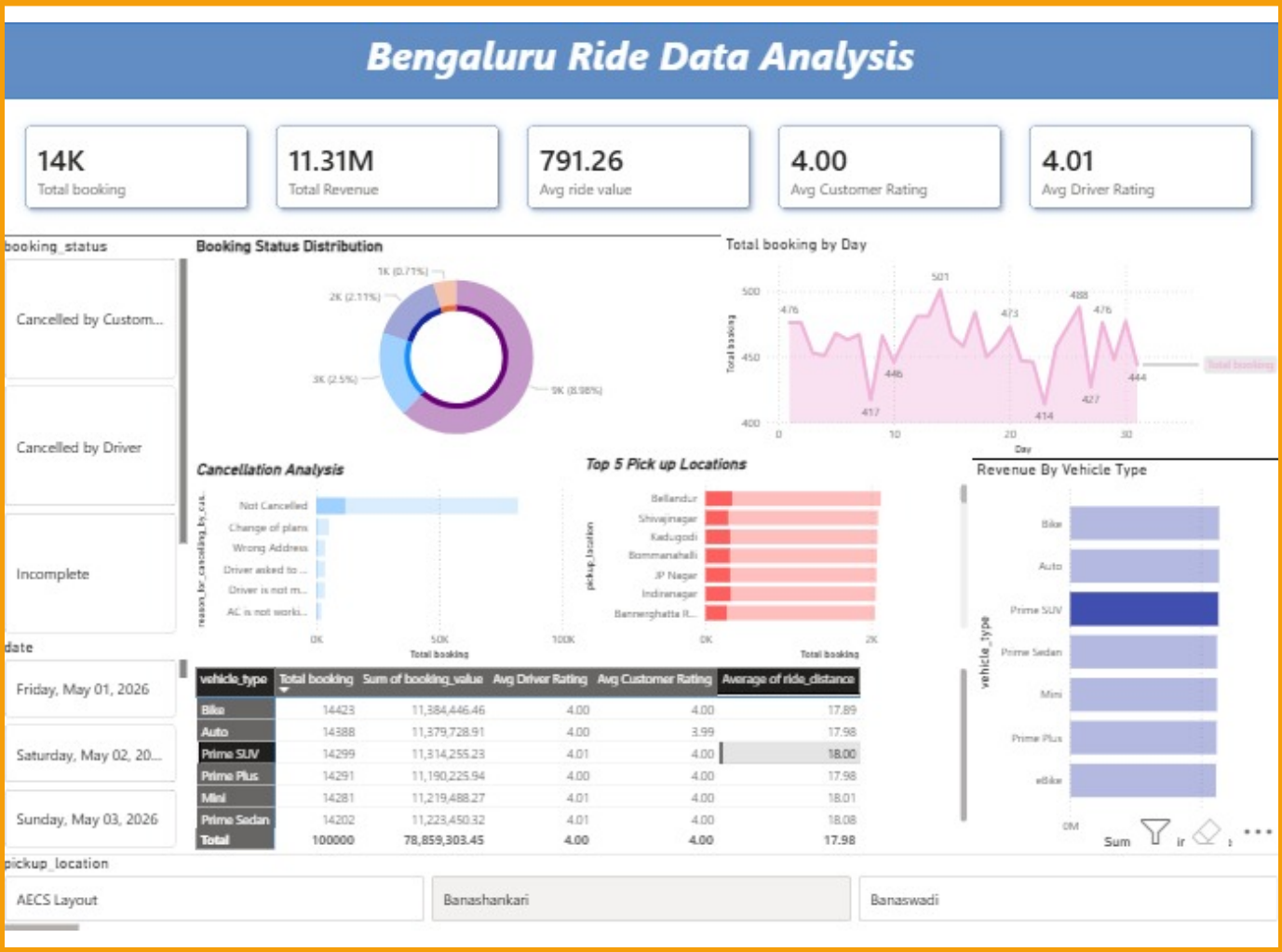


Figure 3 — Power BI interactive dashboard displaying total bookings (14K), total revenue (11.31M), average ride value (791.26), booking status distribution, revenue by vehicle type, top pickup locations, cancellation analysis, and daily booking trends.

Dashboard Highlights

- **14K** total bookings analyzed with full drill-down capability

• **Booking Status Distribution** pie chart with 62% success clearly shown

• **Top 5 Pickup Locations:** Bellandur, Shivajinagar, Kadugodi, JP Nagar, Indiranagar
- **11.31M** total revenue — Power BI calculates in real-time

• **Revenue by Vehicle Type** ranked bar chart — Prime SUV leads

• **Cancellation Analysis** — breakdown of all cancellation reasons

6. Conclusion & Recommendations

This project successfully completed an end-to-end data analytics workflow for a simulated Bengaluru ride booking platform. It demonstrated proficiency in data generation, cleaning, analysis, and visualization, culminating in actionable business insights.

01	Reduce Customer Cancellations Investigate 'Change of plans' and 'Wrong address' patterns. Implement stricter cancellation policies, improve real-time location accuracy, and provide flexible ride modification options.
02	Optimize Driver Experience Address 'Driver not moving towards pickup' issues via improved navigation systems, real-time traffic updates, and incentives for timely pickups.
03	Capitalize on Vehicle Type Performance Leverage higher revenue per ride from Prime Sedan and Prime SUV through promotional campaigns during peak hours. Ensure sufficient Bike availability given their high booking volume.
04	Strategic Location Management Use high-demand zones (Indiranagar, Bellandur, Whitefield) for dynamic pricing, driver allocation, and targeted marketing to maximize efficiency and customer satisfaction.
05	Enhance Service Reliability Deep dive into 'Incomplete Rides' to pinpoint operational or technical failures, aiming to improve ride completion rates and overall service reliability.

Demonstrated Skills

Python	Data loading, cleaning, validation, type conversion, duplicate & missing value handling
SQL	Data filtering, aggregation, grouping, and complex business query solving
Power BI	KPI design, interactive dashboard development, data visualization best practices
ChatGPT	Prompt engineering for synthetic data generation at scale

This project validates an end-to-end Ride Booking Analytics solution — from data generation to strategic insight — using Python, SQL, and Power BI.